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Cold Treatment as a Way to Improve the Wear Resistance of Drilling Tool Material

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Abstract

The paper presents the results of experiments studying the effect of cold treatment (holding samples at -75 °C) on the basic mechanical properties ($\sigma_{0.2}$, σ_B, KCV, HRC) and wear resistance of the drill steels 38XM and 55C2 after quenching from different temperatures. A significant increase in wear resistance (by 18–26%, respectively) was found, with a slight increase in strength (by 1–1.1%) and a decrease in impact toughness (by 1.4–19%) of the samples subjected to cold treatment, compared with materials that underwent conventional heat treatment. It is concluded that cold treatment can be regarded as a promising additional heat-treatment operation for increasing the wear resistance of steels used to manufacture mining working tools – hydraulic breaker picks, drill rods, bit bodies and other mining equipment operating under abrasive wear conditions.

Keywords: *drill steels, cold treatment, heat treatment, wear resistance.*

It is known [1, 2, 3] that if the martensite finish point M_f of a steel lies below room temperature, retained austenite remains in the structure of the material after quenching; its amount increases with the content of carbon and alloying elements in the steel (Fig. 1). The higher their content, the lower the M_f point (Fig. 2).

The presence of retained austenite in the structure of the material of quenched parts adversely affects its physical and mechanical properties, manifesting itself above all in reduced hardness and strength and in deteriorated thermal conductivity and magnetic characteristics. These material properties are of primary importance for mechanical engineering technology. The thermal conductivity of steel has a significant effect on the rate of heat removal from the edge of a cutting tool. It has been established experimentally that the reduced durability of cutters during high-speed cutting on metal-working machine tools is explained by the high temperature arising at the edge, which is caused by the poor thermal conductivity of austenite [1].

[Figure 1 — see original PDF, page 2] Dependence of the amount of retained austenite on the content of alloying elements [2].

Cold treatment, which consists in cooling quenched steel below its M_f temperature, promotes the transformation of retained austenite into martensite, which improves the wear resistance and mechanical properties of the material. In addition, prolonged holding at a temperature well below M_f leads, during subsequent tempering of the steel, to the formation of a large number of very fine carbide particles, called η-carbide, which also contributes to improving the service properties of the

steel [4]. The effect of cold treatment depends on the amount of retained austenite in the quenched steel and increases as the M_f point decreases. Therefore, cold treatment is most effective for carbon steels with a carbon content above 0.6% and for high-alloy steels.

[Figure 2 — see original PDF, page 2] Dependence of the M_f point on carbon content [3].

Table 1. Effect of cold treatment and cryogenic treatment on the wear resistance of foreign tool steels

Steel grade		Cold treatment at -79 °C	Cryogenic treatment at -190 °C
AISI	closest GOST equivalent	improvement in wear resistance, %	
D2	Х12МФ	216	717
S7	6Х4М2ФС	241	403
52100	Х	95	320
O1	9ХВГ	121	218
A10	–	130	164
M1	Р2АМ9К5	45	125
H13	4Х5МФ1С	64	109
M2	Р6М5	17	103
T1	Р18	41	76
P20	38ХМ	23	30
440C	95Х18	28	21

Cold treatment and cryogenic treatment are widely used in engineering as an additional operation after quenching to improve the wear resistance of tool steels [1]. Thus, according to [5] (Table 1), holding quenched samples of a wide range of alloy and high-speed steels used as materials for cutting tools and for cold and hot forming dies leads to a substantial improvement (up to 2.5 and 7 times, respectively) in their wear resistance (the test procedure is not given in [5]).

At the same time, no information on the use of these types of treatment to improve the service properties of mining tool materials subject to abrasive wear has been found in the literature. The present work compares the abrasive wear resistance and mechanical properties of some of the most widely used drill steels after cold treatment with the characteristics of these materials after the heat treatment commonly applied in practice, and discusses the possibility of using the cold treatment operation in mining machine building.

The objects of study were the basic physical and mechanical characteristics ($\sigma_{0.2}$, σ_B, KCV, HRC) and the resistance to abrasive wear (R_w) of steels 38XM and 55C2, as materials for picks of percussive drilling equipment, determined on samples after the heat treatment typical for such steels (oil quenching plus low or medium tempering) and on samples additionally subjected to cold treatment. In the latter case, the samples were placed immediately after quenching in a cryostat with a coolant at -75 °C, where they were held for 4 h.

Experiments to determine the physical and mechanical characteristics were carried out on standard specimens using a Zwick/Roell Z100 tensile testing machine, a Zwick/Roell RKP450 pendulum impact tester and a Zwick/Roell ZHU hardness tester. Wear resistance tests were performed according to ASTM G99 (pin-on-disk method) on cylindrical pin specimens (4 mm in diameter, 14 mm high), which

under a constant load (P = 40 N) were worn for a set time at a speed of 0.257 m/s against an electrocorundum disk. From the results of weighing the specimens before and after exposure to the abrasive, the wear resistance was calculated by the formula:

$$W_R = \rho \cdot P \cdot L / m, N{\cdot}m/mm^3 \quad (1)$$

where ρ is the density of the pin material, g/mm^3; m is the mass loss of the pin as a result of wear, g; L is the wear path (100 m).

The values of the determined material characteristics, as the arithmetic mean of the results of 3 experiments, are presented in Table 2.

Table 2. Results of physical-mechanical tests and wear resistance experiments on samples of drill steels heat-treated without and with cold treatment

Steel grade	Type of treatment	Parameter values				
		yield strength $\sigma_{0.2}$, MPa	ultimate strength σ_B, MPa	impact toughness KCV, J/cm^2	hardness, HRC	wear resistance R_w, N·m/mm^3
38XM	Quenching (from 870 °C) + tempering (200 °C)	1478	1933	40.6	51.1	69.0
38XM	Quenching (from 870 °C) + cold + tempering (200 °C)	1529 (+3.5%)	1953 (+1.0%)	32.9 (-19.0%)	51.6 (+1.0%)	81.9 (+18.6%)
38XM	Quenching (from 950 °C) + tempering (200 °C)	1486	1887	37.5	48.7	–
38XM	Quenching (from 950 °C) + cold + tempering (200 °C)	1486 (0.0%)	1907 (+1.1%)	33.2 (-11.5%)	49.3 (+1.2%)	–
55C2	Quenching (from 870 °C) + tempering (200 °C)	*	*	5.6	59.6	97.0
55C2	Quenching (from 870 °C) + cold + tempering (200 °C)	*	*	4.8 (-14.3%)	60.5 (+1.5%)	122.2 (+25.9%)
55C2	Quenching (from 870 °C) + tempering (360 °C)	*	*	7.1	57.8	–
55C2	Quenching (from 870 °C) + cold + tempering (360 °C)	*	*	7.0 (-1.4%)	58.4 (+1.0%)	–
55C2	Quenching (from 870 °C) + tempering (460 °C)	*	*	22.3	46.9	–
55C2	Quenching (from 870 °C) + cold + tempering (460 °C)	*	*	19.0 (-14.8%)	47.4 (+1.1%)	–

* – brittle fracture of the specimens over a wide range of stresses.

As follows from the data in the table, cold treatment, compared with conventional heat treatment, as in [1, 2, 3], leads to a slight increase in the strength characteristics ($\sigma_{0.2}$, σ_B) and hardness of the tested materials with a simultaneous decrease in impact toughness. (The values of $\sigma_{0.2}$ and σ_B of steel 55C2 could not be determined because of the large scatter in the experimental results, caused, as may be concluded, by embrittlement of the specimens as a result of heat treatment.) Moreover, with increasing austenitizing and tempering temperatures, this effect of cold treatment decreases. Abrasive wear resistance increases by 18 to 26% as a result of exposure to cold.

From the preliminary research results presented, it can be concluded that cold treatment can be regarded as a promising additional heat-treatment operation for increasing the wear resistance of steels used to manufacture mining working tools – hydraulic breaker picks, drill rods, bit bodies and other mining equipment operating under abrasive wear conditions. In doing so, the optimal heat-treatment modes should be found under which the negative effect of cold treatment on the impact toughness of steel will be minimal.

References

1. Клименко А. П. Холод в машиностроении. – М.: Машиностроение, 1969. – С. 247.
2. Чудаков Е. А. Машиностроение. Энциклопедический справочник. Раздел 2. Материалы машиностроения. Т. 3. – М.: ГОНТИ, 1947. – С. 341.
3. Новиков И. И. Теория термической обработки металлов. – М.: Металлургия, 1978. – С. 182.
4. Collins D. N. Deep cryogenic treatment of tool steels: A review // Heat treatment of metals. – 1996. – no 2. – pp. 40–42.
5. Pete Paulin. Technological gear // Gear manufacturing. – 1993. – no 3–4. – pp. 26–29.